# Domain-Adaptive Selective State-Space Learning for Cross-Subject Motor-Imagery EEG Decoding

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**Abstract**

Cross-subject motor-imagery electroencephalography (MI-EEG) decoding is hindered by low signal-to-noise ratios, scarce labeled data, and strong distribution shifts between subjects and sessions. This study proposes SSM-DAN, a domain-adaptive selective state-space network that couples a multi-scale convolutional front-end with bidirectional selective state-space blocks to capture both local and long-range temporal patterns, and reduces subject shift through hierarchical alignment, including a learned source–target normalization layer and label-free test-time adaptation. The framework was evaluated on three public MI-EEG datasets using leave-one-subject-out validation with chronologically separated training, adaptation, and test sessions. The proposed SSM-DAN achieved mean accuracies of 76.12% on BCI Competition IV-2a, 84.09% on BCI Competition IV-2b, and 84.33% on Zhou2016. Ablation studies showed that the contribution of each adaptation component depends on the dataset. These results indicate that combining selective state-space modeling with learned, hierarchical alignment improves cross-subject MI decoding without target labels.

*Keywords:* motor imagery, electroencephalography, brain–computer interface, selective state-space model, Mamba, domain adaptation, test-time adaptation

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## 1. Introduction

Motor-imagery brain–computer interfaces (MI-BCIs) decode movement intentions from non-invasive electroencephalography (EEG) for rehabilitation, communication, and control. Because event-related desynchronization and synchronization signatures are weak, non-stationary, and differ between subjects and sessions [1], a decoder trained on existing users often degrades on a new one, while calibrating every user with labeled trials is time-consuming. This study therefore targets cross-subject MI decoding without target labels, which requires solving two problems together: learning local spectral–spatial and long-range temporal patterns from limited data, and reducing the shift between the training subjects and the new subject.

For the first problem, common spatial pattern (CSP) and filter-bank CSP rely on hand-designed, subject-specific stages that transfer poorly [2, 3]. ConvNets and EEGNet learn filters end to end [4, 5] but reach long-range context only through depth or dilation. Hybrid decoders such as EEG Conformer, ATCNet, and TCFormer add self-attention and dilated temporal convolutional networks (TCNs) [6, 7, 8], but self-attention has quadratic cost [9] and needs more data than EEG datasets usually provide. Selective state-space models (SSMs) instead model long sequences at linear cost [10, 11], and recent MI decoders combine them with convolution [12, 13]; these decoders, however, do not address the shift to a new subject.

For the second problem, domain generalization learns subject-invariant features from source subjects only [14, 15, 16], whereas unsupervised domain adaptation (UDA) also exploits unlabeled target EEG through Riemannian or Euclidean input alignment [17, 18], MK-MMD [19], adversarial learning [20, 21], or class prototypes [22], and test-time adaptation adjusts a trained model on unlabeled test data [23, 24]. Most of these methods act at a single level, yet subject shift appears at several: in the covariance of the raw signals, in the feature statistics inside the network, and in the predictions on later sessions. AutoDIAL [25] addresses the internal level by learning how to mix source and target statistics in normalization layers, but it was proposed for visual domain adaptation.

We therefore propose SSM-DAN, which couples a bidirectional selective state-space encoder with a hierarchical, label-free adaptation scheme in which each stage targets one level of shift: the input signals, the internal features, and the predictions. The contributions are:

- an end-to-end decoder that combines multi-scale convolution, alter-

nating forward/backward selective state-space blocks, and local–global TCN fusion;

- a hierarchical adaptation strategy that reduces subject shift at the input, feature, and prediction levels without target labels; and
- an evaluation on three public MI-EEG datasets with subject-wise accuracy, Cohen's $\kappa$, statistical tests, efficiency analysis, and component ablations.

## 2. Method

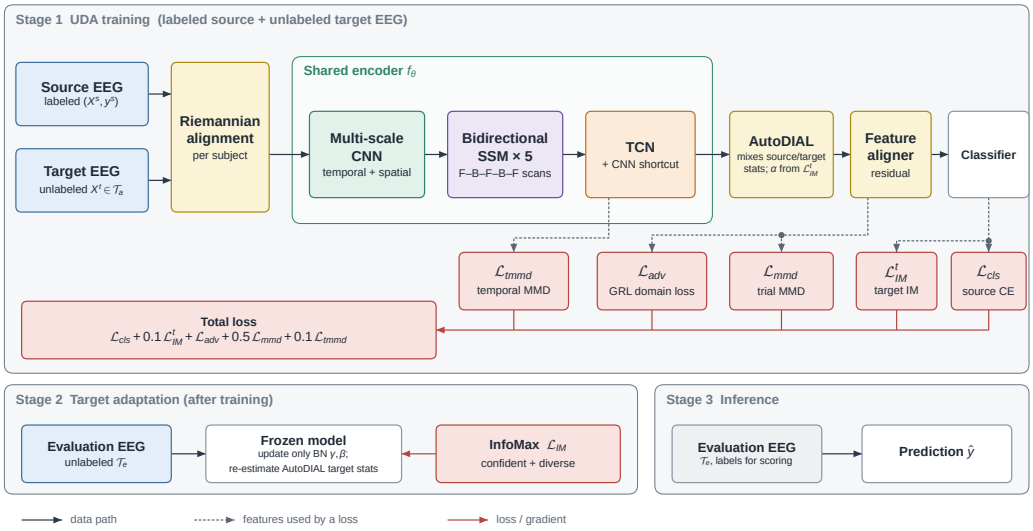



Figure 1: Overview of SSM-DAN: training with labeled source and unlabeled target EEG (Stage 1), label-free test-time adaptation on the unlabeled evaluation sessions $\mathcal{T}_e$ (Stage 2), and prediction (Stage 3).

SSM-DAN has two parts (Fig. 1). An EEG encoder combines multi-scale convolutions for local spectral–spatial features, bidirectional selective state-space blocks for long-range dynamics, and a temporal convolutional fusion head (Sections 2.1–2.3). A hierarchical adaptation scheme reduces subject shift at three levels: Riemannian whitening of the input, AutoDIAL normalization with adversarial and discrepancy losses on the features, and test-time adaptation of the predictions (Sections 2.4–2.7). The model is

trained on labeled source-subject trials and unlabeled target-subject trials (Stage 1), adapted to the unlabeled evaluation trials (Stage 2), and then used for prediction (Stage 3).

### *2.1. Multi-scale convolutional encoder*

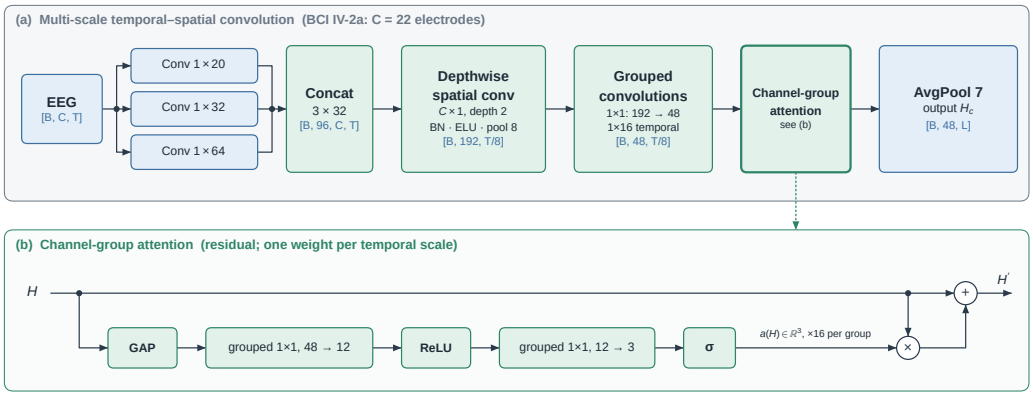



Figure 2: Multi-scale temporal–spatial front-end. (a) Temporal branches, depthwise electrode filtering, grouped refinement, and channel-group attention produce $H_c$. (b) Channel-group attention: one sigmoid weight per temporal scale rescales that scale's 16 channels residually.

Each trial $X \in \mathbb{R}^{C\times T}$ ($C$ electrodes, $T$ samples) passes through three temporal convolutions with kernels of 20, 32, and 64 samples, as in TCFormer [8], followed by depthwise spatial filtering, pooling, and grouped refinement to 16 channels per scale (Fig. 2). A residual channel-group attention reweights the three scales, and a second pooling yields the local feature sequence $H_c \in \mathbb{R}^{48\times L}$.

### 2.2. Bidirectional selective state-space encoder

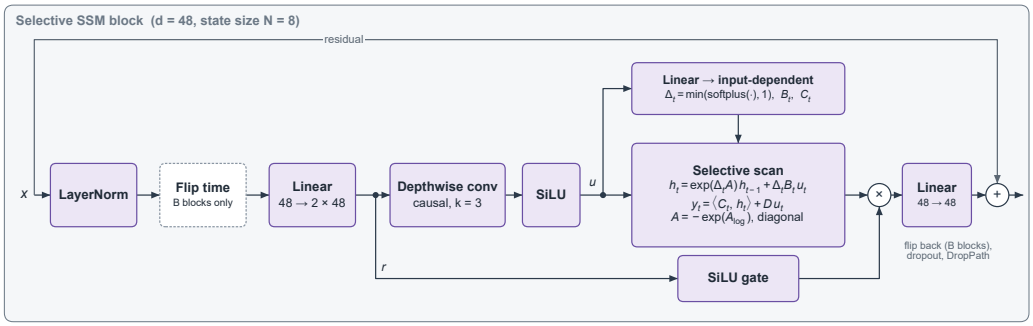



Figure 3: Selective SSM block. Backward blocks reverse and restore the temporal axis around the mixer; forward blocks omit these operations.

To capture trial-wide dependencies, $H_c$ passes through five selective state-space blocks [10, 11], each driving a diagonal recurrence with its projected input $u_t$:

$$h_t = \exp(\Delta_t A) \odot h_{t-1} + \Delta_t B_t \odot u_t, \tag{1}$$

$$z_t = C_t^\top h_t + d \odot u_t, \tag{2}$$

where $A = -\exp(A_{\log})$ is a learned stable decay and $\Delta_t$, $B_t$, and $C_t$ depend on the input, so each block selects which time steps to retain at linear cost. Blocks alternate between forward and backward scanning (Fig. 3), giving bidirectional context without extra blocks. The state dimension is eight, with dropout 0.4 and stochastic depth.

### 2.3. Local–global fusion and classification

The state-space output is reduced to 16 channels and concatenated with the 48-channel local sequence $H_c$, so that the head sees both local and global evidence. Two residual grouped temporal convolutional network (TCN) blocks with kernel length four and dilations one and two refine the resulting 64-channel sequence. The average of its last time step and its temporal mean forms the trial representation $q \in \mathbb{R}^{64}$, made of four 16-channel groups (three convolutional scales and the state-space branch). A norm-constrained grouped classifier produces one set of class scores per group, combined with learned softmax weights.

*2.4. Riemannian signal alignment*

Because EEG covariance structures are shifted between subjects, Riemannian alignment (RA) whitens each subject's trials by the inverse square root of that subject's mean covariance $\bar{R}$, the affine-invariant Riemannian mean of the OAS-shrunk trial covariances [17, 18]:

$$\widetilde{X} = \bar{R}^{-1/2} X. \tag{3}$$

To avoid leakage, $\bar{R}$ is estimated only from data available before evaluation and then applied unchanged: from the training sessions for the validation sessions of each source subject, and from the unlabeled earlier sessions $\mathcal{T}_a$ for the evaluation sessions $\mathcal{T}_e$ of the target subject.

*2.5. Automatic domain alignment layer*

Riemannian whitening aligns input statistics but not shifts arising inside the network. Following AutoDIAL [25], the trial representation $q$ is therefore normalized by a layer that learns how strongly target features are standardized with their own statistics: source features use the source moments $(\mu_s, \sigma_s^2)$, and target features use the moments of a mixture of the source and target $(\mu_t, \sigma_t^2)$ distributions,

$$\mu_\alpha = \alpha\mu_t + (1-\alpha)\mu_s, \tag{4}$$
$$\sigma_\alpha^2 = \alpha\sigma_t^2 + (1-\alpha)\sigma_s^2 + \alpha(1-\alpha)(\mu_t - \mu_s)^2, \tag{5}$$

so that

$$\hat{q}^s = \gamma \odot \frac{q^s - \mu_s}{\sqrt{\sigma_s^2 + \epsilon}} + \beta, \qquad \hat{q}^t = \gamma \odot \frac{q^t - \mu_\alpha}{\sqrt{\sigma_\alpha^2 + \epsilon}} + \beta, \tag{6}$$

with shared affine parameters $\gamma$ and $\beta$. The mixing weight $\alpha \in (0, 1)$, initialized to 0.5 and learned separately for each of the four feature groups, interpolates between source statistics ($\alpha \to 0$) and target statistics ($\alpha \to 1$). Moments are computed per mini-batch during training and tracked as running averages (momentum 0.1), which are mixed with the learned $\alpha$ at inference; source validation trials always use source statistics.

*2.6. Training objective*

A residual $64 \to 128 \to 64$ aligner precedes the classifier and alignment losses. Besides the source cross-entropy $\mathcal{L}_{\text{cls}}$, Stage 1 uses a gradient-reversal

domain loss $\mathcal{L}_{\text{adv}}$ [20], MK-MMD losses on trial representations ($\mathcal{L}_{\text{mmd}}$) and TCN sequences ($\mathcal{L}_{\text{tmmd}}$) [19], and the information-maximization loss [23]

$$\mathcal{L}_{\text{IM}} = \frac{1}{N}\sum_{j=1}^{N} H(p_j) - H\left(\frac{1}{N}\sum_{j=1}^{N} p_j\right), \tag{7}$$

where $p_j$ is the predicted class distribution of the $j$-th of $N$ unlabeled trials and $H(\cdot)$ the entropy. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + 0.1\mathcal{L}^{t}_{\text{IM}} + \mathcal{L}_{\text{adv}} + 0.5\mathcal{L}_{\text{mmd}} + 0.1\mathcal{L}_{\text{tmmd}}. \tag{8}$$

where $\mathcal{L}^{t}_{\text{IM}}$ is computed on target trials from $\mathcal{T}_a$. The alignment losses see the AutoDIAL output with $\alpha$ fixed, so $\alpha$ is learned only through $\mathcal{L}^{t}_{\text{IM}}$.

### *2.7. Test-time adaptation*

The checkpoint with the highest source-validation accuracy is adapted to the unlabeled evaluation trials $\mathcal{T}_e$ by information-maximization test-time adaptation (IM-TTA): only the BatchNorm scales and shifts are updated, which limits overfitting to a small target set [24], by minimizing Eq. (7) over $\mathcal{T}_e$ for five Adam passes (learning rate $10^{-4}$). The AutoDIAL target moments are then re-estimated in one pass over the same trials, with $\alpha$ and the source moments fixed. The labels of $\mathcal{T}_e$ are used only for scoring.

## 3. Experimental Setup

### *3.1. Datasets*

Three public MI-EEG datasets, accessed through MOABB/Braindecode [26], are used (Table 1).

#### *3.1.1. BCI Competition IV-2a*

Nine participants were recorded in two sessions on different days, each with 288 trials of left-hand, right-hand, both-feet, and tongue imagery, from 22 EEG electrodes at 250 Hz (0.5–100 Hz band-pass, 50-Hz notch; the three EOG channels are not used) [28]. Participant sex and age are not reported.

Table 1: Summary of the MI-EEG datasets (EOG channels excluded).

| Dataset | Subjects | Sex (M/F) | Channels | Sessions | Classes |
|---|---|---|---|---|---|
| BCI IV-2a | 9 | n.r. | 22 | 2 | left hand, right hand, feet, tongue |
| BCI IV-2b | 9 | n.r.[a] | 3 | 5 | left hand, right hand |
| Zhou2016 | 4 | 1/3 | 14 | 3 | left hand, right hand, feet |

n.r.: not reported in the dataset description. [a]The nine participants were drawn from a study of ten (6/4, 24.7 ± 3.3 years) [27]; the split of the nine released participants is not reported.

#### *3.1.2. BCI Competition IV-2b*

Nine right-handed participants, drawn from a study of ten (six male, four female; $24.7 \pm 3.3$ years) [27], performed left- versus right-hand imagery recorded from three bipolar derivations around C3, Cz, and C4 at 250 Hz (the three EOG channels are not used). Sessions 1–3 form the calibration portion and sessions 4–5 the evaluation portion [28]; this chronological split is kept, and the dataset tests whether the method remains useful with only three electrodes.

#### *3.1.3. Zhou2016*

Four healthy participants (one male, three female; 22–28 years; sex reported only at group level) were each recorded in three sessions separated by days to months [29]. Each session has two runs of 75 trials (25 per class: left hand, right hand, feet) from 14 electrodes at 250 Hz; each trial has a 5-s imagery cue followed by a 4-s rest.

### *3.2. Chronological LOSO protocol with test-time adaptation*

Each leave-one-subject-out (LOSO) fold holds out one target subject and uses the others as labeled sources, with the session split in Table 2. Earlier target sessions are used without labels in Stage 1; the evaluation sessions are excluded from training and model selection, and their unlabeled EEG is used only by IM-TTA before scoring.

This gives nine folds for BCI IV-2a and IV-2b and four for Zhou2016, with identical hyperparameters in every fold; the checkpoint with the highest source-validation accuracy is used. For Zhou2016, an all-session target score is reported as an auxiliary diagnostic.

Table 2: Chronological session allocation in each LOSO fold. Evaluation-session labels are used only for scoring.

| Dataset | Source subjects (labeled) | | Target subject | |
|---|---|---|---|---|
| | Training | Validation | Stage 1 UDA (unlabeled) | IM-TTA + evaluation (labels withheld) |
| BCI IV-2a | session 1 | session 2 | session 1 | session 2 |
| BCI IV-2b | sessions 1–3 | sessions 4–5 | sessions 1–3 | sessions 4–5 |
| Zhou2016 | sessions 1–2 | session 3 | sessions 1–2 | session 3 |

### *3.3. Preprocessing and augmentation*

EEG channels are resampled to 250 Hz and converted to microvolts. BCI IV-2a uses the full MOABB event window and BCI IV-2b drops its last 0.5 s, both unfiltered; Zhou2016 is band-pass filtered at 8–30 Hz and cropped to 0.5–5.0 s after cue onset.

Z-score statistics are fitted on the source-training partition. For BCI IV-2a and IV-2b, labeled source mini-batches are doubled with the segmentation and reconstruction (S&R) augmentation [30] used by TCFormer [8], which recombines eight-segment pieces of same-class trials; Zhou2016 uses no augmentation.

### *3.4. Implementation details*

All experiments run in Kaggle notebooks on one NVIDIA Tesla T4 GPU (16 GB). BCI IV-2a and IV-2b use a batch size of 48, at most 125 epochs, and Adam (learning rate $9 \times 10^{-4}$, $\beta_1 = 0.5$, weight decay $10^{-3}$); Zhou2016 uses a batch size of 32, 100 epochs, a learning rate of $10^{-3}$, and weight decay $10^{-4}$. A three-epoch linear warm-up is followed by cosine decay. The seed (0), data split, and initialization are shared by all methods and variants, so each comparison differs only in the evaluated component.

### *3.5. Comparison methods*

Baselines are grouped by target-data access, each group being compared with the SSM-DAN variant of equal access. Source-only decoders (EEG Conformer [6], ATCNet [7], TCFormer [8], EEGEncoder [31], SATrans-Net [32]) are compared with SSM-DAN; the UDA method RMETNet [33], which also sees the unlabeled earlier target sessions, with SSM-DAN (Stage 1 UDA),

i.e., without IM-TTA; and test-time adaptation methods (T-TIME [34], BFT [35]) with SSM-DAN (IM-TTA), i.e., without Stage 1 adaptation. Every baseline was re-trained from its authors' released implementation under the protocol of Section 3.2: the same LOSO folds, chronological session split (Table 2), per-dataset preprocessing, random seed, and checkpoint selection on the labeled source-validation sessions. Evaluation-session labels never enter training, model selection, or hyperparameter choice, and the test-time adaptation methods receive the evaluation EEG only as an unlabeled stream. Architecture, loss, optimizer, and batch size follow each method's original recommendations. Results reported under other protocols are therefore not copied into Tables 3–5; they are contrasted with the re-runs in Section 4.5. For RMETNet on Zhou2016, a ridge of $10^{-3}$ times the mean eigenvalue is added to each trial covariance to keep it positive definite.

### *3.6. Evaluation measures and reporting*

Accuracy is reported per target subject, and accuracy and Cohen's $\kappa$ as mean $\pm$ standard deviation across folds. Because every evaluation set is balanced across classes, $\kappa$ is a linear rescaling of accuracy; two complementary measures are therefore added. The information transfer rate (ITR) of Wolpaw et al. [36] expresses accuracy $P$ with $N$ classes as communication speed,

$$\mathrm{ITR} = \frac{60}{T}\left[\log_2 N + P\log_2 P + (1-P)\log_2\frac{1-P}{N-1}\right] \ \text{bits/min}, \qquad (9)$$

where $T$ is the length of the decoded EEG window (4 s for BCI IV-2a and IV-2b and 4.5 s for Zhou2016); it is computed per target subject, set to zero at or below chance, and then averaged. Class-wise behaviour is summarised by the recall of each class and the macro-averaged F1 score, which reveal whether accuracy is concentrated on a subset of classes. Paired accuracy differences use a two-sided Wilcoxon signed-rank test with exact $p$-values. Because each method is compared with the SSM-DAN variant of the same target-data access (Section 3.5), the $p$-values are adjusted for multiple comparisons by the Holm–Bonferroni procedure within each of these three families (eight, one, and two tests), and a difference is called significant when the adjusted $p_{\mathrm{Holm}} < 0.05$. Efficiency is measured by trainable parameters, mean Stage 1 training time per fold on the T4, and batch-one CPU response time (mean of 500 passes after 100 warm-up passes), excluding IM-TTA, preprocessing, and data transfer.

To interpret the dataset-dependent RA ablation, a model-free geometric diagnostic is also applied. OAS trial covariances from the labeled source subjects form one affine-invariant Riemannian centroid $M_k$ per class, and each held-out target evaluation trial $C_i$ is assigned to its nearest source centroid. Label selectivity is measured by the true-versus-nearest-wrong margin

$$m_i = \min_{k \neq y_i} d_R(C_i, M_k) - d_R(C_i, M_{y_i}), \tag{10}$$

where a positive change after RA means that the target trial moves toward its correct source class relative to every competing class. For comparability across channel and class configurations, the subject-level margin change is divided by the mean pre-RA separation between source-class centroids. Controlled BCI IV-2a variants retain either C3/Cz/C4, left/right trials, or both. The four primary margin comparisons use exact two-sided permutation tests between datasets or paired Wilcoxon tests within BCI IV-2a, followed by Holm correction. Target labels enter only this retrospective diagnostic and never training, adaptation, or model selection.

## 4. Results

Tables 3–5 report subject-wise chronological LOSO results. Standard deviations describe variation across target subjects, not repeated seeds.

### *4.1. BCI Competition IV-2a*

Table 3 reports the four-class results for the nine BCI IV-2a targets.

SSM-DAN obtained the highest accuracy and $\kappa$, and significantly outperformed every source-only baseline and its own source-only variant on all nine targets ($p_{\text{Holm}} = 0.0312$). IM-TTA improved every target over SSM-DAN (Stage 1 UDA). In the UDA group, SSM-DAN (Stage 1 UDA) significantly exceeded RMETNet ($60.11 \pm 11.05\%$) on all nine targets. In the test-time adaptation group, SSM-DAN (IM-TTA) significantly exceeded BFT on all nine targets and exceeded T-TIME on seven, although the latter difference was not significant.

### *4.2. BCI Competition IV-2b*

Table 4 reports the two-class results for the nine BCI IV-2b targets, which provide only three bipolar EEG derivations.

Table 3: Per-subject LOSO accuracy (%), with mean Cohen's $\kappa$ and mean information transfer rate (ITR, Eq. (9)), on BCI IV-2a (four classes). $p$: two-sided exact Wilcoxon signed-rank test against the SSM-DAN model of the same category (Section 3.5); $p_{\text{Holm}}$: $p$ adjusted by the Holm–Bonferroni procedure within each category (Section 3.6). –: reference model.

| Method | Target subject accuracy (%) | | | | | | | | | Mean ± SD | | | $p$ | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | S1 | S2 | S3 | S4 | S5 | S6 | S7 | S8 | S9 | Acc. (%) | $\kappa$ | ITR (bits/min) | | |
| *Source-only training* | | | | | | | | | | | | | | |
| Conformer [6] | 58.68 | 43.06 | 78.82 | 43.06 | 44.44 | 41.67 | 72.92 | 65.28 | 49.31 | 55.25 ± 13.39 | 0.403 ± 0.179 | 5.29 ± 4.58 | 0.0039 | 0.0312 |
| ATCNet [7] | 69.44 | 40.97 | 76.74 | 50.35 | 43.40 | 42.01 | 61.81 | 70.49 | 57.99 | 57.02 ± 12.74 | 0.427 ± 0.170 | 5.73 ± 4.19 | 0.0039 | 0.0312 |
| TCFormer [8] | 65.28 | 49.65 | 79.86 | 54.86 | 61.81 | 38.89 | 71.88 | 66.67 | 58.33 | 60.80 ± 11.49 | 0.477 ± 0.153 | 6.80 ± 4.02 | 0.0039 | 0.0312 |
| EEGEncoder [31] | 64.24 | 49.31 | 78.82 | 46.18 | 54.86 | 40.97 | 74.31 | 65.97 | 47.92 | 58.06 ± 12.55 | 0.441 ± 0.167 | 6.04 ± 4.43 | 0.0039 | 0.0312 |
| SATrans-Net [32] | 68.06 | 47.92 | 73.96 | 53.47 | 61.11 | 45.14 | 73.61 | 71.53 | 63.19 | 62.00 ± 10.34 | 0.493 ± 0.138 | 7.09 ± 3.62 | 0.0039 | 0.0312 |
| SSM-DAN (source-only) | 68.06 | 51.04 | 86.46 | 61.11 | 64.24 | 39.58 | 72.92 | 73.61 | 62.15 | 64.35 ± 12.77 | 0.525 ± 0.170 | 8.22 ± 4.97 | 0.0039 | 0.0312 |
| *Unsupervised domain adaptation* | | | | | | | | | | | | | | |
| RMETNet [33] | 67.01 | 39.24 | 77.78 | 55.90 | 49.65 | 52.78 | 63.54 | 66.67 | 68.40 | 60.11 ± 11.05 | 0.468 ± 0.147 | 6.52 ± 3.73 | 0.0039 | 0.0039 |
| SSM-DAN (Stage 1 UDA) | 79.17 | 47.57 | 92.71 | 73.61 | 57.64 | 59.72 | 82.29 | 82.64 | 73.96 | 72.15 ± 13.58 | 0.629 ± 0.181 | 11.60 ± 6.31 | 0.0039 | 0.0312 |
| *Test-time adaptation* | | | | | | | | | | | | | | |
| T-TIME [34] | 67.01 | 53.47 | 78.47 | 64.58 | 68.75 | 49.65 | 79.51 | 69.44 | 65.28 | 66.24 ± 9.34 | 0.550 ± 0.125 | 8.56 ± 3.73 | 0.0742 | 0.0742 |
| BFT [35] | 64.24 | 44.44 | 76.74 | 65.62 | 61.81 | 51.39 | 73.61 | 70.49 | 59.38 | 63.08 ± 9.77 | 0.508 ± 0.130 | 7.42 ± 3.51 | 0.0039 | 0.0078 |
| SSM-DAN (IM-TTA) | 77.43 | 47.92 | 88.19 | 66.32 | 64.24 | 61.81 | 79.86 | 82.99 | 73.61 | 71.37 ± 11.78 | 0.618 ± 0.157 | 10.98 ± 5.27 | 0.0039 | 0.0312 |
| **SSM-DAN** | **79.51** | **52.43** | **93.40** | **73.96** | **69.44** | **65.97** | **88.89** | **86.11** | **75.35** | **76.12 ± 11.96** | **0.682 ± 0.159** | **13.30 ± 6.15** | – | – |

Table 4: Per-subject LOSO accuracy (%), with mean Cohen's $\kappa$ and mean information transfer rate (ITR, Eq. (9)), on BCI IV-2b (two classes). $p$: two-sided exact Wilcoxon signed-rank test against the SSM-DAN model of the same category (Section 3.5); $p_{\text{Holm}}$: $p$ adjusted by the Holm–Bonferroni procedure within each category (Section 3.6). –: reference model.

| Method | Target subject accuracy (%) | | | | | | | | | Mean ± SD | | | $p$ | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | S1 | S2 | S3 | S4 | S5 | S6 | S7 | S8 | S9 | Acc. (%) | $\kappa$ | ITR (bits/min) | | |
| *Source-only training* | | | | | | | | | | | | | | |
| Conformer [6] | 75.31 | 62.86 | 55.62 | 92.50 | 94.69 | 74.06 | 82.81 | 82.50 | 81.88 | 78.03 ± 12.02 | 0.561 ± 0.240 | 4.55 ± 3.51 | 0.0156 | 0.1094 |
| ATCNet [7] | 76.25 | 65.71 | 57.19 | 90.00 | 96.88 | 77.50 | 84.06 | 84.69 | 82.19 | 79.38 ± 11.40 | 0.588 ± 0.228 | 4.89 ± 3.58 | 0.0078 | 0.0625 |
| TCFormer [8] | 78.44 | 69.64 | 64.38 | 90.62 | 95.00 | 75.94 | 86.87 | 85.31 | 84.38 | 81.18 ± 9.37 | 0.624 ± 0.187 | 5.17 ± 3.16 | 0.0820 | 0.3281 |
| EEGEncoder [31] | 73.44 | 65.71 | 59.06 | 92.50 | 97.19 | 85.00 | 85.94 | 86.56 | 83.75 | 81.02 ± 11.76 | 0.620 ± 0.235 | 5.48 ± 3.81 | 0.0547 | 0.2734 |
| SATrans-Net [32] | 77.50 | 69.29 | 69.69 | 90.62 | 97.50 | 79.69 | 87.50 | 84.38 | 79.69 | 81.76 ± 8.78 | 0.635 ± 0.176 | 5.36 ± 3.45 | 0.2109 | 0.6094 |
| SSM-DAN (source-only) | 77.81 | 68.57 | 63.13 | 89.38 | 96.56 | 73.44 | 84.69 | 86.25 | 85.00 | 80.54 ± 10.05 | 0.611 ± 0.201 | 5.07 ± 3.44 | 0.0156 | 0.1094 |
| *Unsupervised domain adaptation* | | | | | | | | | | | | | | |
| RMETNet [33] | 73.12 | 57.86 | 55.00 | 91.56 | 94.06 | 82.81 | 85.62 | 88.75 | 83.13 | 79.10 ± 13.39 | 0.582 ± 0.268 | 5.04 ± 3.54 | 0.0117 | 0.0117 |
| SSM-DAN (Stage 1 UDA) | 78.44 | 70.71 | 63.44 | 90.31 | 97.81 | 83.75 | 89.69 | 95.63 | 85.94 | 83.97 ± 10.71 | 0.679 ± 0.214 | **6.42 ± 3.98** | 1.0000 | 1.0000 |
| *Test-time adaptation* | | | | | | | | | | | | | | |
| T-TIME [34] | 59.69 | 65.36 | 85.31 | 93.75 | 97.19 | 85.00 | 74.69 | 91.25 | 59.06 | 79.03 ± 13.94 | 0.581 ± 0.279 | 5.24 ± 4.39 | 0.3125 | 0.3125 |
| BFT [35] | 75.00 | 65.00 | 59.69 | 93.44 | 94.69 | 79.06 | 88.12 | 94.37 | 81.25 | 81.18 ± 12.12 | 0.624 ± 0.242 | 5.60 ± 3.96 | 0.0078 | 0.0156 |
| SSM-DAN (IM-TTA) | 77.81 | 68.57 | 66.25 | 93.75 | 96.25 | 81.88 | 89.69 | 94.69 | 81.25 | 83.35 ± 10.50 | 0.667 ± 0.210 | 6.15 ± 3.92 | 0.2031 | 0.6094 |
| **SSM-DAN** | 77.81 | 69.29 | 65.94 | 90.00 | 97.19 | 84.69 | 91.56 | 95.31 | 85.00 | **84.09 ± 10.42** | **0.682 ± 0.208** | **6.42 ± 3.90** | – | – |

SSM-DAN again obtained the highest mean accuracy. It exceeded its source-only variant on all seven targets with non-tied accuracies, ATCNet on all eight, and EEG Conformer on eight of nine ($p = 0.0078$–$0.0156$), but these differences did not remain significant after Holm correction ($p_{\text{Holm}} = 0.0625$–$0.1094$); its advantages over TCFormer, EEGEncoder, SATrans-Net, and SSM-DAN (Stage 1 UDA) were small and not significant, with IM-TTA helping on only three targets. Within the matched groups, SSM-DAN (Stage 1 UDA) significantly exceeded RMETNet and SSM-DAN (IM-TTA) significantly exceeded BFT; SSM-DAN (IM-TTA) also exceeded T-TIME ($79.03 \pm 13.94\%$) on average, but not significantly, as T-TIME varied far more across targets.

### *4.3. Zhou2016*

Table 5 reports the three-class results on target session 3 of Zhou2016. With four folds, the smallest attainable two-sided Wilcoxon $p$-value is 0.125, so no difference can be significant.

Table 5: Per-subject LOSO accuracy (%), with mean Cohen's $\kappa$ and mean information transfer rate (ITR, Eq. (9)), on session 3 of Zhou2016 (three classes). Bold: best per column. $p$: two-sided exact Wilcoxon signed-rank test against the SSM-DAN model of the same category (Section 3.5); $p_{\text{Holm}}$: $p$ adjusted by the Holm–Bonferroni procedure within each category (Section 3.6). –: reference model.

| Method | Target subject accuracy (%) | | | | Mean ± SD | | | $p$ | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|---|---|---|---|
| | S1 | S2 | S3 | S4 | Acc. (%) | $\kappa$ | ITR (bits/min) | | |
| *Source-only training* | | | | | | | | | |
| Conformer [6] | 76.00 | 79.33 | 70.67 | 82.67 | $77.17 \pm 4.43$ | $0.657 \pm 0.066$ | $7.86 \pm 1.86$ | 0.1250 | 1.0000 |
| ATCNet [7] | 72.00 | 82.67 | 76.67 | 74.67 | $76.50 \pm 3.93$ | $0.648 \pm 0.059$ | $7.60 \pm 1.70$ | 0.1250 | 1.0000 |
| TCFormer [8] | 68.00 | 78.67 | 66.67 | 80.67 | $73.50 \pm 6.22$ | $0.603 \pm 0.093$ | $6.67 \pm 2.39$ | 0.1250 | 1.0000 |
| EEGEncoder [31] | 62.00 | 84.00 | 74.67 | 76.67 | $74.33 \pm 7.92$ | $0.615 \pm 0.119$ | $7.07 \pm 2.98$ | 0.2500 | 1.0000 |
| SATrans-Net [32] | 81.33 | 80.00 | 67.33 | 70.00 | $74.67 \pm 6.09$ | $0.620 \pm 0.091$ | $7.06 \pm 2.40$ | 0.1250 | 1.0000 |
| SSM-DAN (source-only) | 72.00 | 78.67 | 71.33 | 78.67 | $75.17 \pm 3.51$ | $0.627 \pm 0.053$ | $7.10 \pm 1.41$ | 0.1250 | 1.0000 |
| *Unsupervised domain adaptation* | | | | | | | | | |
| RMETNet [33] | 70.00 | **89.33** | 80.67 | 84.67 | $81.17 \pm 7.14$ | $0.718 \pm 0.107$ | $9.63 \pm 3.29$ | 0.8750 | 1.0000 |
| SSM-DAN (Stage 1 UDA) | 79.33 | 84.67 | **87.33** | 78.00 | $82.33 \pm 3.82$ | $0.735 \pm 0.057$ | $9.90 \pm 1.92$ | 0.6250 | 1.0000 |
| *Test-time adaptation* | | | | | | | | | |
| T-TIME [34] | **88.67** | 84.00 | 81.33 | **86.00** | $\mathbf{85.00 \pm 2.69}$ | $\mathbf{0.775 \pm 0.040}$ | $\mathbf{11.06 \pm 1.46}$ | 0.5000 | 0.8750 |
| BFT [35] | 84.00 | 86.67 | 76.00 | 84.67 | $82.83 \pm 4.07$ | $0.743 \pm 0.061$ | $10.13 \pm 1.94$ | 0.8750 | 1.0000 |
| SSM-DAN (IM-TTA) | 79.33 | 84.00 | 83.33 | 83.33 | $82.50 \pm 1.85$ | $0.738 \pm 0.028$ | $9.90 \pm 0.89$ | 0.2500 | 1.0000 |
| **SSM-DAN** | 84.00 | 83.33 | 84.67 | 85.33 | $84.33 \pm 0.75$ | $0.765 \pm 0.011$ | $10.70 \pm 0.39$ | – | – |

SSM-DAN had the most consistent performance across targets and exceeded every source-only baseline and its source-only variant on all four tar-

gets, except EEGEncoder, which it exceeded on three. In the UDA group, RMETNet was slightly below its reference SSM-DAN (Stage 1 UDA) on average and far more variable across targets, although it obtained the highest accuracy on target 2. In the test-time adaptation group, T-TIME obtained the highest mean accuracy on this dataset, above both its reference SSM-DAN (IM-TTA) and the complete SSM-DAN, while BFT was on par with the reference. SSM-DAN also remained above TCFormer in the auxiliary score pooled over all three target sessions.

### *4.4. Information transfer rate and class-wise performance*

In practical terms, the accuracy gains of SSM-DAN translate into faster communication. On BCI IV-2a, adaptation raised its ITR by about 60% over the source-only variant (13.3 versus 8.2 bits/min; Table 3), so a new user could issue markedly more commands per minute without any labeled calibration. On BCI IV-2b and Zhou2016 the adaptive methods operated in a similar range, mirroring the accuracy results.

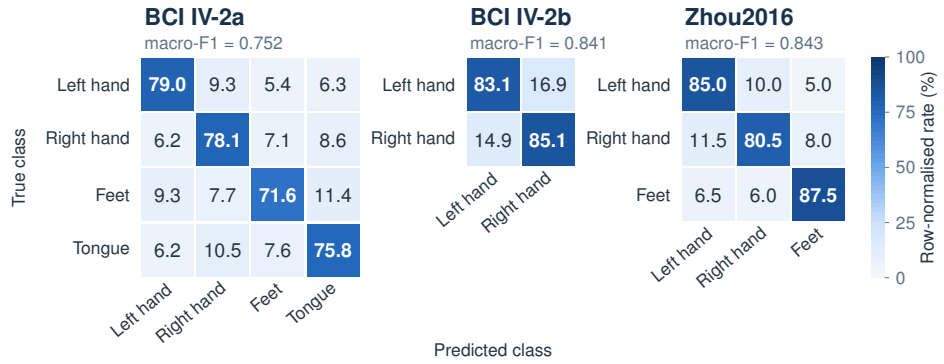



Figure 4: Mean confusion matrices of SSM-DAN on the evaluation sessions of BCI IV-2a (left), BCI IV-2b (middle), and Zhou2016 (right). Each subject's matrix is normalized by the number of trials of each true class and then averaged over target subjects, so the diagonal gives the mean recall of each class; the macro-averaged F1 score is given under each dataset name.

Fig. 4 shows that these gains are spread across classes rather than obtained by favoring an easy one: macro-F1 tracks accuracy on every dataset and no class is systematically neglected. This matters for test-time adaptation, whose entropy objective could otherwise collapse predictions onto a few classes. The remaining errors are also physiologically plausible: on BCI

IV-2a, feet imagery was most often confused with tongue imagery, two tasks represented near the midline of the sensorimotor cortex, whereas the lateralized hand classes were recognized best.

### *4.5. Comparison with published results*

Table 6 contrasts the cross-subject accuracies reported in the original articles with the re-runs of this study. The published values are not directly comparable, because the protocols differ in four respects. (i) *Task*: the T-TIME and BFT articles report two-class left- versus right-hand MI, whereas BCI IV-2a and Zhou2016 are evaluated here with four and three classes. (ii) *Session shift and target data*: several protocols train and test within the same recording session, and the EEGEncoder article reports no cross-subject result at all, its multi-subject figure being obtained with the test subjects' own first sessions in the training set; here every target is unseen during training and scored on a later session recorded on a different day. (iii) *Model selection*: the released scripts of SATrans-Net and RMETNet report the highest, or the mean of the five highest, per-epoch accuracies on the held-out subject, i.e., the evaluation labels select the epoch; here the checkpoint is chosen on source data only. (iv) *Transductive access*: RMETNet uses the held-out evaluation trials themselves as the target domain of its MMD loss and Riemannian alignment, whereas here domain adaptation sees only the earlier, unlabeled target sessions. Each of (ii)–(iv) makes a protocol easier or its estimate more optimistic, which largely explains why most re-runs are lower on BCI IV-2a. Where the split matches ours, the gaps are small: the TCFormer benchmark [8] uses the same session split and differs only in averaging the last-epoch model over five seeds and in model-specific preprocessing, and the re-runs of TCFormer and ATCNet lie within three points of it on both BCI datasets, while EEG Conformer is higher here. On BCI IV-2b, where the shift between calibration and evaluation sessions is smaller, the re-runs reach or exceed several published values (e.g., SATrans-Net, 81.76 vs. 79.64%).

### *4.6. Ablation study*

Table 7 follows the primary chronological protocol. The variant without domain adaptation removes every Stage 1 domain-adaptation component (AutoDIAL, target information maximization, and the adversarial and both MMD losses) while retaining Riemannian alignment and IM-TTA; all variants otherwise use the same folds and checkpoint rule.

Table 6: Published cross-subject accuracy (%) of the baselines versus their re-runs under the protocol of this study (Table 2). –: not reported; settings (a)–(e) are given below the table.

| | | Published | | | This study (chronological LOSO) | | |
|---|---|---|---|---|---|---|---|
| Method | Setting | BCI IV-2a | BCI IV-2b | Zhou2016 | BCI IV-2a | BCI IV-2b | Zhou2016 |
| EEG Conformer [6] | (a) [8] | 45.59 | 75.25 | – | 55.25 | 78.03 | 77.17 |
| ATCNet [7] | (a) [8] | 59.66 | 80.94 | – | 57.02 | 79.38 | 76.50 |
| TCFormer [8] | (a) | 63.00 | 81.34 | – | 60.80 | 81.18 | 73.50 |
| EEGEncoder [31] | (b) | – | – | – | 58.06 | 81.02 | 74.33 |
| SATrans-Net [32] | (c) | 68.13 | 79.64 | – | 62.00 | 81.76 | 74.67 |
| RMETNet [33] | (d) | 71.39 | 80.93 | – | 60.11 | 79.10 | 81.17 |
| T-TIME [34] | (e) | 79.30$^{\dagger}$ | – | – | 66.24 | 79.03 | 85.00 |
| BFT [35] | (e) | 77.47$^{\dagger}$ | – | 84.38$^{\dagger}$ | 63.08 | 81.18 | 82.83 |

(a) Same session split as here, last-epoch model averaged over five seeds, model-specific preprocessing. (b) No cross-subject result; the 74.48% on BCI IV-2a trains on session 1 of all subjects, including the test subject. (c) LOSO without a reported session split, best per-epoch test accuracy. (d) One session per subject, held-out subject used as MMD/RA target, mean of the five best epochs. (e) Online test-time adaptation on session 1 only. $^{\dagger}$Two-class (left- vs. right-hand) task.

Table 7: Ablation results: mean ± standard deviation of accuracy (%) and Cohen's $\kappa$ across target subjects.

| Dataset | Metric | SSM-DAN | Without Riemannian alignment | Without domain adaptation | Without test-time adaptation |
|---|---|---|---|---|---|
| BCI IV-2a | Acc. | $\mathbf{76.12 \pm 11.96}$ | $70.29 \pm 11.60$ | $71.37 \pm 11.78$ | $72.15 \pm 13.58$ |
| | $\kappa$ | $\mathbf{0.682 \pm 0.159}$ | $0.604 \pm 0.155$ | $0.618 \pm 0.157$ | $0.629 \pm 0.181$ |
| BCI IV-2b | Acc. | $\mathbf{84.09 \pm 10.42}$ | $83.58 \pm 10.46$ | $83.35 \pm 10.50$ | $83.97 \pm 10.71$ |
| | $\kappa$ | $\mathbf{0.682 \pm 0.208}$ | $0.672 \pm 0.209$ | $0.667 \pm 0.210$ | $0.679 \pm 0.214$ |
| Zhou2016 | Acc. | $\mathbf{84.33 \pm 0.75}$ | $84.00 \pm 3.16$ | $82.50 \pm 1.85$ | $82.33 \pm 3.82$ |
| | $\kappa$ | $\mathbf{0.765 \pm 0.011}$ | $0.760 \pm 0.047$ | $0.738 \pm 0.028$ | $0.735 \pm 0.057$ |

On BCI IV-2a, removing Riemannian alignment, Stage 1 domain adaptation, or test-time adaptation each significantly reduced accuracy. On BCI IV-2b and Zhou2016, the complete model remained the best variant, but its margins were small and not significant. The contribution of each component is therefore dataset dependent.

### *4.7. Why Riemannian alignment helps BCI IV-2a*

The geometric diagnostic distinguishes generic domain contraction from alignment that is useful for classification (Fig. 5). In the original tasks, nearest-centroid accuracy increased from 30.32% to 42.52% on BCI IV-2a (+12.19 percentage points; exact paired Wilcoxon $p = 0.0039$), but only from 56.66% to 57.63% on BCI IV-2b (+0.97 points; $p = 0.4961$). More importantly, the normalized decision-margin gain was 0.1517 on four-class BCI IV-2a versus 0.0287 on binary BCI IV-2b (exact permutation $p_{\text{Holm}} = 0.0389$). The effect remained when BCI IV-2a was reduced to C3/Cz/C4 (0.2264; $p_{\text{Holm}} = 0.0110$ versus BCI IV-2b), ruling out channel count as the sole explanation.

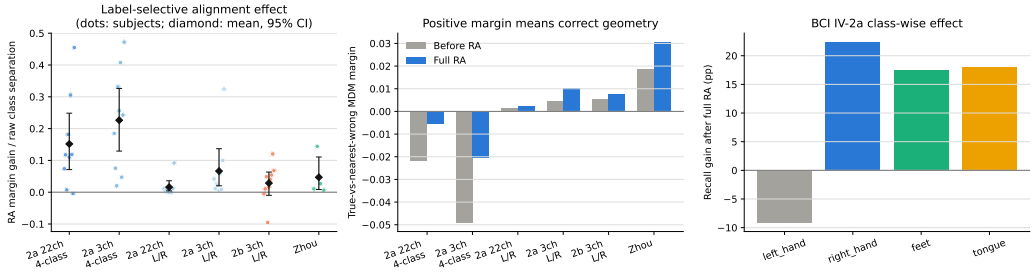



Figure 5: Label-selective effect of RA. Left: subject-level gain in the true-versus-nearest-wrong Riemannian margin (Eq. 10), normalized by pre-RA source-class separation; diamonds and bars show the mean and bootstrap 95% confidence interval. Middle: mean raw margin before and after full RA, where positive values indicate correct class geometry. Right: class-wise change in diagnostic MDM recall on the original 22-channel BCI IV-2a task. This model-free diagnostic is separate from SSM-DAN accuracy.

Within BCI IV-2a, restricting the task to left/right reduced the normalized margin gain from 0.1517 to 0.0160 at 22 channels ($p_{\text{Holm}} = 0.0389$) and from 0.2264 to 0.0665 at three channels ($p_{\text{Holm}} = 0.0234$). Full RA particularly recovered classes with poor unaligned transfer: diagnostic recall increased by 22.38, 17.44, and 18.06 points for right hand, feet, and

tongue, respectively, while left-hand recall decreased by 9.10 points. Conversely, after matching both the three-channel montage and left/right task, the margin-gain difference between BCI IV-2a and IV-2b was not significant ($p = 0.4174$). Thus, the evidence points to the richer class-conditional covariance geometry of the original four-class BCI IV-2a task, rather than the number of electrodes or the magnitude of marginal shift alone. This diagnostic explains the direction of the neural ablation but is not itself an SSM-DAN accuracy result.

### *4.8. Computational efficiency*

Table 8 reports trainable parameters, Stage 1 training time, and CPU response time. The source-only variant is smaller because it omits the AutoDIAL layer, aligner, and domain discriminator; RA and IM-TTA do not change parameter count.

Stage 1 adaptation accounts for most of the extra training time of SSM-DAN over its source-only variant (nearly doubling it on Zhou2016), whereas IM-TTA adds little. SSM-DAN has a parameter count comparable to ATCNet and TCFormer and far below EEG Conformer, while T-TIME and BFT are the smallest and fastest methods.

Table 8: Computational cost: trainable parameters, mean Stage 1 training time per fold, and single-trial CPU response time. SSM-DAN variants as in Section 3.5.

| Method | BCI IV-2a | | | BCI IV-2b | | | Zhou2016 | | |
|---|---|---|---|---|---|---|---|---|---|
| | Params (K) | Train (min) | CPU (ms) | Params (K) | Train (min) | CPU (ms) | Params (K) | Train (min) | CPU (ms) |
| *Source-only training* | | | | | | | | | |
| Conformer | 789.6 | 11.14 | 11.45 | 759.1 | 4.93 | 9.34 | 858.7 | 1.53 | 10.34 |
| ATCNet | 113.7 | 48.58 | 13.33 | 112.8 | 60.19 | 11.35 | 113.3 | 8.09 | 11.91 |
| TCFormer | 127.2 | 38.37 | 14.96 | 123.4 | 31.04 | 11.17 | 125.6 | 5.89 | 15.97 |
| EEGEncoder | 197.2 | 13.49 | 26.51 | 196.2 | 14.98 | 26.88 | 196.8 | 3.64 | 25.68 |
| SATrans-Net | 155.6 | 10.19 | 12.78 | 153.7 | 4.99 | 11.10 | 155.0 | 1.32 | 11.38 |
| SSM-DAN (source-only) | 98.4 | 47.01 | 25.08 | 94.6 | 36.10 | 13.68 | 96.8 | 6.29 | 13.36 |
| *Unsupervised domain adaptation* | | | | | | | | | |
| RMETNet | 193.8 | 45.88 | 11.15 | 159.1 | 21.96 | 4.90 | 306.6 | 23.74 | 7.19 |
| SSM-DAN (Stage 1 UDA) | 132.0 | 58.45 | 18.44 | 128.2 | 48.08 | 13.35 | 130.4 | 11.42 | 14.65 |
| *Test-time adaptation* | | | | | | | | | |
| T-TIME | 9.5 | 5.85 | 7.71 | 6.3 | 3.07 | 5.08 | 8.4 | 1.34 | 6.72 |
| BFT | 40.6 | 1.47 | 3.08 | 39.9 | 0.96 | 2.61 | 51.0 | 0.36 | 2.84 |
| SSM-DAN (IM-TTA) | 131.9 | 40.73 | 22.12 | 128.1 | 33.21 | 14.25 | 130.3 | 6.46 | 18.41 |
| SSM-DAN (w/o RA) | 132.0 | 58.41 | 21.63 | 128.2 | 47.95 | 13.33 | 130.4 | 11.66 | 18.88 |
| **SSM-DAN** | **132.0** | **59.82** | **22.20** | **128.2** | **54.21** | **17.40** | **130.4** | **11.70** | **19.14** |

## 5. Discussion

The benefit of each adaptation component is dataset dependent: it was clear on BCI IV-2a but small and not significant on BCI IV-2b and Zhou2016, where repeated-seed experiments are needed to confirm the differences. The geometric analysis suggests that this difference is determined by label selectivity rather than by how strongly RA contracts the domains: full RA reduced class-conditional covariance shift on BCI IV-2b, yet produced little improvement in the correct-versus-wrong class margin. On BCI IV-2a, the effect survived reduction to three channels but weakened substantially when the four-class task was reduced to left/right. RA therefore appears particularly useful for the heterogeneous lateral and midline covariance patterns of this task; the results do not imply that increasing the number of classes will generally improve RA.

The alternating scans exploit complete offline trials and are non-causal, so online use would require forward-only processing.

Because IM-TTA uses unlabeled evaluation-session signals, the method is transductive and must be compared with baselines of identical target access. Further limitations are the three-channel BCI IV-2b setting, the small Zhou2016 cohort, incomplete matched baselines, and the lack of sex-stratified analysis, as participant sex is reported only in aggregate or not at all.

## 6. Conclusion

We presented SSM-DAN, a cross-subject MI-EEG framework that combines multi-scale convolution and bidirectional selective state-space modeling with hierarchical UDA and constrained IM-TTA, evaluated under a chronological LOSO protocol that keeps target labels out of training, adaptation, and model selection. It reached mean accuracies of 76.12% on BCI IV-2a, 84.09% on BCI IV-2b, and 84.33% on Zhou2016. Matched baselines and repeated-seed experiments are still needed before drawing broader conclusions.

## CRediT authorship contribution statement

**[Author name]:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.


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## Data availability

All EEG data analyzed in this study are publicly available. BCI Competition IV datasets 2a and 2b [28] and the Zhou2016 dataset [29] were accessed through the MOABB library [26]; no new data were collected.

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